

REVIEW

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# How do generative artificial intelligence (AI) tools and large language models (LLMs) influence language learners' critical thinking in EFL education? A systematic review

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## Abstract

As generative artificial intelligence (AI) tools and large language models (LLMs)-powered applications develop rapidly in the era of algorithms, it should be integrated thoughtfully to enhance English as a Foreign Language (EFL) teaching and learning without replacing learners' critical thinking (CT). This study systematically analyzes the impact of generative AI tools and LLMs on language learners' CT in EFL education using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework to identify, evaluate, and synthesize relevant studies from 2022 to 2025. A thorough review of 15 selected studies focuses on generative AI tools and LLMs' dual nature, research methods, main focuses, theory and models, limitations and challenges, and future directions in the field based on Web of Science (WoS), SCOPUS, ERIC, ProQuest, and Google Scholar. The findings identified generative AI tools and LLMs possessed both the potential to nurture and the risk of hindering CT in EFL education. 66.67% of studies reported generative AI tools and LLMs' positive role in CT, while 33.33% of studies reported its negative role in CT. Furthermore, 3 types of research methods, 3 key themes of research focus, and 4 groups of theoretical perspectives were examined. However, 4 kinds of limitations in this field remain, including research scope, user dependency, generative AI reliability, and pedagogical integration. Future research can focus on assessing long-term effects, broadening research scope, promoting responsible AI use, and refining pedagogical strategies. Finally, Limitations, implications and future direction of this study were discussed.

**Keywords:** Generative AI, Large language models, Critical thinking, EFL education, Systematic review

## Introduction

Artificial intelligence (AI) has advanced rapidly nowadays, bringing significant changes to education. One of the most notable developments is generative artificial intelligence (AI), which can create text, images, and even simulate conversations (Cao et al., 2023; Gozalo-Brizuela & Garrido-Merchan, 2023). A breakthrough in this field was the release of ChatGPT by OpenAI in November 2022 (Cao et al., 2023). Built on Large Language

Models (LLMs), including GPT-3.5, GPT-4, and GPT-4o, ChatGPT can generate human-like responses, translate languages, summarize information, and assist with writing and learning tasks (Aydın & Karaarslan, 2023; Azaria, 2023; Orrù et al., 2023). Since then, other different LLMs-driven generative AI tools, like DeepSeek and Claude, have also emerged, offering almost similar capabilities.

With the rise of these AI-generative technologies, many countries have started to explore their role in education. China's Ministry of Education announced in 2023 that it would promote AI-assisted learning while ensuring that students develop independent thinking skills (Chen & Gong, 2025). Meanwhile, the European Union has introduced regulations under its AI Act to ensure the ethical use of AI in education (Nikolinakos, 2023). Other countries have also taken steps to integrate AI responsibly into classrooms, with initiatives to provide teachers with guidance on using AI tools effectively.

Generative AI and LLMs-powered tools have quickly become valuable for language learning. They help students practice writing by offering instant feedback, enhance speaking skills through interactive conversations, and improve reading comprehension by simplifying complex texts (Jegede, 2024). In some schools, teachers use AI-generated exercises and personalized language learning plans to support students' progress (Melisa et al., 2025).

Currently, generative AI is transforming the way students learn English by enhancing writing, speaking, reading, and personalized learning experiences. Many studies have extensively examined the impact of generative AI and LLMs on EFL teaching and learning. Studies showed that generative AI tools significantly enhanced students' vocabulary, grammar, and writing skills in English language courses (Al Mahmud, 2023; Song & Song, 2023). Tools like ChatGPT, helped EFL learners to practice conversations, receive instant grammar corrections, and generate essay drafts (Amyatun & Kholis, 2023; Waziana et al., 2024). AI-powered chatbots improved students to practice speaking English by having conversations, helping them build confidence and fluency (Jinming & Daniel, 2024). For reading practice, generative AI can summarize long articles, explain difficult words, and create comprehension questions (Hidayat, 2024). Many learning platforms, like Duolingo and ELSA Speak, used AI to personalize lessons, adjusting to each student's needs and progress (Betaubun et al., 2023; Shinde, 2024).

Teachers are also finding generative AI and LLMs useful in EFL classrooms. Some used AI-generative tools to grade assignments and give feedback, saving time and allowing them to focus on teaching (Jegede, 2024). AI-powered feedback can be more detailed and personalized than traditional teacher feedback, helping students identify and correct mistakes more efficiently. In some schools, AI-powered assistants help answer students' questions outside of class, giving extra support when teachers are not available (Chen & Lu, 2023; Srinivasa et al., 2022).

However, while these studies highlighted the benefits of generative AI in improving English language skills, they focused mostly on technical aspects: grammar accuracy, fluency, and vocabulary expansion, fewer studies explore its impact on deeper cognitive abilities like critical thinking (CT) in EFL context. Learning English language is not just about using correct grammar and expanding vocabulary, it also involves analyzing texts, forming arguments, and thinking independently (Mogea, 2022). It is universally recognized as a key skill of 21st-century, CT skills play a crucial role in the development of

students' cognitive abilities and their ability to analyze, evaluate, and synthesize information (Kuloğlu & Karabekmez, 2022). According to Elder and Paul (2020), CT skill enables learners to tackle complex problems with clarity, create innovative solutions, and make informed decisions based on logic and evidence. In EFL courses, CT ability is essential for tasks: writing essays, analyzing texts, and engaging in discussions. When students write, they need to organize ideas, evaluate sources, and present arguments logically (Fitria, 2023). In reading comprehension, they must identify main ideas, detect bias, and question the reliability of information. During discussions, students practice expressing opinions, defending viewpoints, and considering different perspectives. These skills are at the core of English language learning, helping EFL students not just use English fluently but also think deeply to develop cognitive skills.

Another research gap is the lack of long-term studies on how generative AI and LLMs influence EFL students' thinking habits. Marzuki et al. (2023) and Tseng and Lin (2024) focused on short-term improvements in their writing and comprehension, but they did not explore whether students who used AI-generative tools regularly develop stronger or weaker CT skills over time. Additionally, while Girdhar and Khachatryan (2023) discussed how generative AI helped with creative writing and discussion exercises, there is little research on how students critically engage with AI-generated content. For instance, do students question the accuracy of AI responses, or do they simply accept them as correct? Do AI-generated texts encourage deeper analysis, or do they lead to passive learning? These questions remain largely unexplored.

With generative AI and LLMs-based tools providing instant answers and corrections, there is a risk that students may rely too much on AI-generated content instead of thinking for themselves. Aymen and Zakarya (2024) and Liu and Wang (2024) even worried that generative AI may make students too dependent on automated responses, reducing their ability to analyze, question, and express their own ideas in EFL courses. A student writing an essay might use AI chatbots to generate ideas, structure the paper, and even rewrite sentences (Datskiv et al., 2024). While this can improve writing quality, it may also reduce the need for independent critical thinking. Instead of struggling with how to express an idea, students might simply accept whatever generative AI suggests, without critically evaluating its accuracy or relevance (Aymen & Zakarya, 2024). This can lead to surface-level learning, where students completed tasks but did not deeply engage with the English language. In English reading and research, generative AI can summarize texts and highlight key points, making it easier to understand difficult material. However, true comprehension requires more than just summarization (Yilin et al., 2023). Students must learn to ask questions, connect ideas, and assess the credibility of sources. If they rely too much on AI-generated summaries, they may miss out on developing analytical reading skills.

When facing these challenges, EFL educators need to find a balanced approach. Instead of replacing traditional learning methods, generative AI and LLMs-powered tools should be used as supporting tools that encourages deeper analytical thinking (Hading et al., 2024). In the future, the role of generative AI and LLMs in EFL education will continue to grow. While it offers exciting possibilities for EFL learning, it is important to ensure that it enhances rather than weakens critical thinking. Through using generative AI and LLMs-powered tools thoughtfully, students can gain the benefits of technology

while still developing their ability to analyze, question, and express ideas independently, which are essential not only in EFL teaching and learning but in all aspects of life (Dat-skiv et al., 2024). Thus, research on the influence of generative AI and LLMs for fostering language learners' critical thinking in EFL is a meaningful and imperative topic.

Current literature reviews (see Table 1) focused on the role of generative AI and LLMs-powered tools, especially ChatGPT-type tools, in education, in a wide range of language learning and critical thinking, instead of focusing on EFL teaching and learning. A review by Fitria (2023) examined how ChatGPT helped students write English essays. Fitria (2023) described how users can access ChatGPT, input prompts, and receive structured essay responses. Furthermore, a integrative literature review by Joseph (2023) discussed how LLM-based tools like ChatGPT can be integrated into language teaching to improve CT and cognitive development. Joseph (2023) introduced a framework for "Critical AI Literacy" to help teachers use generative AI effectively while addressing its limitations. Moreover, a systematic review by Melisa et al. (2025) focused on generative AI's impact on CT skills in higher education. It reviewed past research and found that ChatGPT can help students develop analytical skills by providing different perspectives, assisting with argument construction, and supporting information analysis. Additionally, a systematic review by Sardi et al. (2025) explored how generative AI influenced students' self-regulated learning (SRL) and CT skills. Using a systematic review method, Sardi et al. (2025) analyzed studies published between 2022 and 2024, showing that Generative AI tools, like ChatGPT, played a positive role in supporting SRL and CT by providing personalized feedback and adaptive learning experiences. Overall, the four literature reviews provided useful insights into how generative AI and LLMs-powered tools influence students' CT, but they also showed gaps that they did not focus on EFL education, which need further investigation.

While 4 literature reviews have examined the role of generative AI and LLMs in education and CT, most of these studies adopt a general educational lens. These studies lack a specific focus on EFL teaching and learning. The EFL context poses unique challenges that differ from English or second-language learning environments, especially when it comes to integrating critical thinking into language instruction. Moreover, current literature has largely emphasized the technical benefits of generative AI, such as grammar correction, vocabulary expansion, and writing assistance, while overlooking its potential impact on higher-order cognitive skills like CT in EFL learners (Fitria, 2023). There is also a lack of investigation into the long-term influence of generative AI tools on language learners' thinking habits and their engagement with AI-generated content.

Therefore, this systematic review addresses existing research gaps by examining how generative AI and LLMs influence CT in EFL education. It aims to provide a focused and nuanced understanding of both the opportunities and cognitive risks associated with their use in EFL contexts.

Building on previous findings, this review specifically explores the dual nature of generative AI and LLMs, the research methods adopted, main research focuses, underlying theories and models, current limitations and challenges, as well as future directions. Unlike prior reviews, it offers a deeper analysis of how generative AI tools and LLMs shape CT development in EFL learners and provides practical insights for EFL educators, curriculum designers, and policymakers. The goal is to support the creation of

**Table 1** The overview of the reviews of generative AI and LLMs in education or on CT

| Author (Year)        | Time Scope | Scope                                                                                                                                  | Methods                       | Data resource                                                                                                         | Findings                                                                                                                                                                                                                                                      | Gaps                                                                                                                                                                                                                 |
|----------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fitria (2023)        | 2022–2023  | Use of ChatGPT in English essay writing for educational purposes                                                                       | Critical review               | Twitter posts, direct interaction with ChatGPT                                                                        | ChatGPT can generate structured English essays, uses active/passive voice, appropriate tenses, and follows logical writing order                                                                                                                              | Lack of focus on critical thinking development and no assessment of AI's impact on learners' cognitive skills in EFL contexts                                                                                        |
| Joseph (2023)        | 2018–2023  | Synthesizing with UNESCO's Education for Sustainable Development (ESD) competencies to propose a Critical AI Literacy (CAIL) framework | Integrative literature review | 17 Peer-reviewed journal articles from Google Scholar and ScienceDirect, focusing on CDL in English language teaching | Identified studies proposing CDL frameworks aligned with ESD cognitive, affective, and conative domains. Proposed a CAIL framework for integrating LLMs like ChatGPT into language teaching to develop critical thinking and sustainable cognitive structures | The review does not directly address how generative AI tools and LLMs influence language learners' critical thinking in EFL education. It focuses on proposing a framework on critical thinking                      |
| Sardi et al. (2025)  | 2022–2024  | Impact of generative AI (e.g., ChatGPT) on self-regulated learning (SRL) and critical thinking (CT) in education                       | Systematic review             | 38 peer-reviewed articles                                                                                             | 71.4% of studies reported positive AI impact on SRL; 62.5% reported positive impact on CT; concerns about over-reliance and loss of autonomy                                                                                                                  | The review does not specifically focus on language learners or EFL education, nor does it isolate critical thinking in language learning contexts                                                                    |
| Melisa et al. (2025) | 2023–2024  | Higher education students across various academic disciplines                                                                          | Systematic review             | 19 peer-reviewed articles                                                                                             | ChatGPT can enhance critical thinking by providing diverse perspectives and aiding analysis, but over-reliance may hinder self-reflection and evaluation skills                                                                                               | The review does not specifically focus on EFL education or language learners. It also lacks detailed exploration of LLMs beyond ChatGPT and their specific impact on critical thinking in language learning contexts |

effective, evidence-based generative AI-enhanced learning strategies that foster independent, analytical and critical thinking in line with the demands of the digital age.

## **Literature review**

### **What is critical thinking?**

Critical thinking (CT) is the ability to think clearly, logically, and independently. It involves questioning ideas, analyzing information, and making reasoned judgments. According to Ennis (2018), critical thinking is “reasonable reflective thinking that is focused on deciding what to believe or do.” Elder and Paul (2020) described CT as “the art of analyzing and evaluating thinking with a view to improving it.” Briefly, CT helps individuals make better decisions by considering different perspectives and carefully examining evidence. There are several key components of CT skills. Analysis allows individuals to break down complex ideas into smaller parts to understand them better (Hart et al., 2021). Reasoning helps connect different pieces of information logically to form sound conclusions (Kuhn, 2018). Evaluation enables people to assess the quality and reliability of information before accepting it (Alsaleh, 2020). Explanation and elaboration involve clearly expressing ideas and the reasoning behind them (Van Peppen et al., 2018). Self-regulation helps individuals reflect on their own thinking, identify biases, and improve their judgment (Arkadiy, 2021). These skills work together to help learners think in a structured and rational way.

In language learning, particularly in English as a Foreign Language (EFL) education, CT plays an essential role, especially in writing, reading, and speaking. In English writing course, students need to organize their ideas logically, construct strong arguments, and consider opposing viewpoints (Kim et al., 2025). In English reading, CT ability helps learners analyze texts, recognize biases, and differentiate facts from opinions (Al Roomy, 2022; Yamin et al., 2022). In English speaking, it enables learners to express their thoughts clearly, support their ideas with evidence, and engage in meaningful discussions (Jocuns, 2021). Thus, through developing CT skills, English language learners become more independent thinkers who can communicate effectively and evaluate information critically.

### **How are generative AI tools and LLMs used in EFL teaching and learning?**

Generative artificial intelligence (AI) refers to technology that can create new content, including text, images, and audio, based on patterns it has learned from large amounts of data (Banh & Strobel, 2023). Large language models (LLMs) are a type of generative AI designed specifically for processing and generating human-like text (Parasuraman, 2024). These models, such as ChatGPT and DeepSeek, can understand, generate, and refine language, making them useful tools for EFL teaching and learning.

Generative AI and LLMs-powered tools are widely used to support different aspects of EFL education. One significant advantage is that LLMs-driven generative AI tools provide instant feedback and personalized learning experiences, helping EFL learners to study at their own pace (Jegade, 2024). Furthermore, in English writing courses, generative AI can help generate ideas, suggest sentence structures, rewrite sentences for clarity, and correct grammar mistakes (Al Mahmud, 2023). It is helpful for learners to improve their writing style and accuracy. In English reading courses, generative AI can



summarize long texts, highlight key points, and provide explanations of difficult words and phrases (Hidayat, 2024). Liu and Wang (2024) noted that some generative AI tools also guide learners in analyzing texts critically by suggesting different perspectives on a topic. For EFL speaking practice, AI-powered chatbots can simulate conversations and provide instant feedback on pronunciation, fluency, and grammar (Jinming & Daniel, 2024). Learners can practice speaking English in a low-pressure environment, improving their confidence. In English language listening training, generative AI can generate custom audio materials based on a language learner's level, interests, and learning goals (Betaubun et al., 2023; Shinde, 2024), which may help EFL learners develop better listening skills by exposing them to a variety of accents, speech patterns, and topics.

While generative AI and LLMs-powered tools offer many benefits in EFL classroom, it also has limitations. Over-reliance on generative AI and LLMs can reduce independent thinking and problem-solving skills (Hading et al., 2024). Additionally, AI-generated content may sometimes contain errors and biases, so EFL learners need to critically evaluate the information they receive (Datskiv et al., 2024). Despite these concerns, LLMs-powered generative AI tools remain a powerful tool that, when used correctly, can greatly support EFL learning and development.

#### **How do generative AI and LLMs influence EFL learners' critical thinking?**

Generative AI and LLMs can have both positive and negative effects on EFL learners' critical thinking. When used effectively, generative AI can help learners develop important thinking skills (Datskiv et al., 2024). One major benefit is that generative AI encourages critical reading by presenting multiple perspectives on a topic (Aymen & Zakarya, 2024). When learners interact with AI-generated discussions, they are exposed to different viewpoints, which helps them question ideas and analyze arguments more deeply (Aymen & Zakarya, 2024). Generative AI and LLMs can also improve logical reasoning by providing feedback on reasoning processes (Jegade, 2024). When learners ask generative AI to explain a concept or support an argument, they can evaluate the logic behind the response and refine their own thinking. Additionally, generative AI and LLMs-powered tools support reflective learning by offering instant feedback on writing, speaking, and problem-solving (Nabilla et al., 2024), which helps EFL learners identify their mistakes, adjust their approach, and improve their CT skills over time.

However, generative AI and LLMs-powered tools in EFL education also have potential drawbacks that may weaken CT skills if not used carefully. One major concern is over-reliance on generative AI and LLMs-powered tools which can reduce EFL learners' ability to think independently (Rizkiani et al., 2024). If EFL learners constantly depend on generative AI-type tools to generate ideas, correct errors, or analyze texts, they may struggle to develop their own reasoning and problem-solving abilities. Another issue is that AI-generated content may contain biases or errors, which can mislead learners if they do not critically evaluate the information (Datskiv et al., 2024). Since generative AI models are trained on large datasets, they sometimes unknowingly reinforce stereotypes or present inaccurate facts. Lastly, generative AI and LLMs-powered tools might oversimplify complex thinking cognitive processes, especially in CT (Datskiv et al., 2024). If learners use LLMs-powered generative AI applications to generate essays and arguments

without actively engaging in the CT process, they may miss the opportunity to develop deeper analytical skills.

Consequently, generative AI and LLMs can be a valuable tool for enhancing EFL learners' CT, but it should be used carefully. EFL learners need to remain active participants in their cognitive learning process, questioning AI-generated content, verifying information, and using generative AI and LLMs-powered tools as support rather than replacement for independent thinking.

### Research objectives

This study aims to explore whether generative artificial intelligence (AI) tools and large language models (LLMs) kill or nurture language learners' critical thinking in English as a Foreign Language (EFL) teaching and learning to present a systematic literature review of relevant studies. Based on the research objectives, there are six research questions (RQ) as follows:

- RQ1: Do generative AI tools and LLMs kill or nurture critical thinking in EFL education?
- RQ2: What research methods were employed in generative AI Tools and LLMs for fostering critical thinking in EFL education?
- RQ3: What were the main research focuses of generative AI Tools and LLMs for fostering critical thinking in EFL education?
- RQ4: What were theory or models involved in generative AI Tools and LLMs for fostering critical thinking in EFL education?
- RQ5: What were the limitations or challenges identified in generative AI Tools and LLMs for fostering critical thinking in EFL education?
- RQ6: What are the future directions for generative AI Tools and LLMs for fostering critical thinking in EFL education?

### Research methodology

This systematic review is guided by the preferred reporting items for systematic reviews and meta-analyses' (PRISMA) statement (Page et al., 2021). The methodology for this systematic review is designed to ensure a comprehensive and unbiased collection, analysis, and synthesis of literature concerning the studies of generative artificial intelligence (AI) tools and large language models (LLMs) on learners' critical thinking (CT) in English as a Foreign Language (EFL) education.

### Search strategy

A search method was developed to retrieve relevant materials from the Web of Science (WoS), Scopus, ProQuest, ERIC, and Google Scholar databases from their establishment until 28 February, 2025. The first four databases were chosen for their extensive scope and established reputation in educational and interdisciplinary research. Google Scholar was used as a complementary source to capture potentially relevant but less formally indexed studies, especially those published in open-access or practitioner-oriented venues. The search keywords were carefully constructed to reflect the intersection of



generative artificial intelligence (AI), critical thinking (CT), and English language education. The terms of search query are follows: (“Generative AI\*” OR “AI-assisted learning” OR “large language models” OR “LLMs” OR “chatbots for learning” OR “AI-generated content”) AND (“critical thinking” OR “think critically” OR “critical reasoning”) AND (“English education” OR “English language teaching” OR “TESOL” OR “EFL” OR “ESL”). The combination of these terms using Boolean operators ensures both breadth and specificity, allowing for the identification of the most relevant literature at the convergence of these domains.

### Eligibility criteria and study selection

The PRISMA process was employed to refine the results. The review targeted publications from 2022 to 2025, deliberately selected to ensure relevance and timeliness as ChatGPT, the first typical LLMs-powered generative AI tool, began gaining broad adoption in 2022. As shown in Table 2, well-defined inclusion criteria were applied to ensure the relevance and quality of the selected studies. Firstly, the target population was teachers and students in an EFL setting. Secondly, only publications written in English were considered for inclusion. Furthermore, the scope was limited to the fields of generative AI and LLMs on CT. Additionally, due to the novelty of this topic and the limited availability of relevant literature, suitable journals, dissertations, and conference papers were carefully selected. Studies that failed to meet these criteria, such as those review papers, book chapters, presentations and policy reports or without empirical evidence or those centered on unrelated generative AI or LLMs applications, were excluded. This rigorous selection process ensured that the review’s conclusions were based on reliable and relevant research.

### Study selection

During the initial screening, five databases yielded a total of 3243 studies, and 4 additional records from other sources were found. After applying exclusion and inclusion criteria and using Endnote’s Find Duplicates feature, 3006 studies were deleted. After that, a total of 241 studies for title and abstract were screened and 45 studies were found, which were then reviewed in full text for eligibility. A total of 28 articles were excluded for different reasons: 4 studies were not focusing on CT, 10 studies were not in EFL education setting, 10 studies were not studying in generative AI or LLMs, and 4 studies

**Table 2** Inclusion and exclusion criteria applied to the review

| Type        | Inclusion                                                                           | Exclusion                                                                       |
|-------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Time        | January 1, 2022 to February 28, 2025                                                | Any other points in time                                                        |
| Setting     | EFL setting                                                                         | Any other settings                                                              |
| Scope       | Generative AI and LLMs on CT in EFL education                                       | Any other EFL education form without Generative AI and LLMs on CT               |
| Publication | Peer & Non-peer reviewed journal articles, dissertations and Conference proceedings | Review papers, Book chapters, Presentations, Policy reports and any other types |
| Language    | Written in English                                                                  | Written in any other languages                                                  |
| Acquisition | Full text is available                                                              | No full text is available                                                       |

were review papers. Finally, 15 studies were obtained and examined for coding. Detailed information on each fully reviewed and included paper was recorded in a table, including the authors, publication year, regions, study design, key focuses and limitations (see Appendix 1). A PRISMA approach was employed to conduct an in-depth analysis of these 15 articles, with the detailed process outlined in Fig. 1.

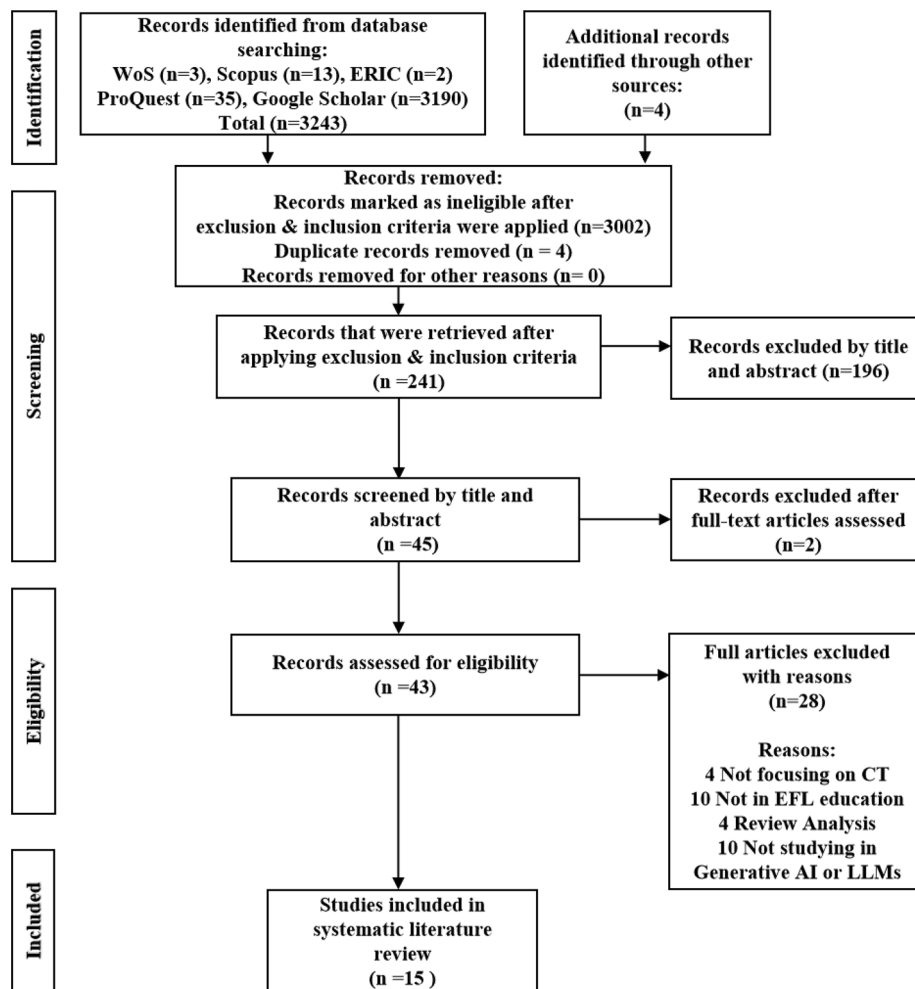
Duplicate articles were identified by cross-checking search results across databases, focusing on titles, publication years, and author names. The titles and abstracts of all retrieved papers were independently screened by two authors based on the selection criteria. Study eligibility was documented using the PRISMA flow diagram. The authors independently selected studies that passed the initial screening and retrieved their full texts. Data extraction was conducted in duplicate by two authors for each included publication after a thorough full-text review. The consensus was reached through mutual agreement based on a comparison of the authors' records. In cases of disagreement, a third author was consulted to facilitate resolution.

### Quality assessment

To ensure methodological rigor and minimize the risk of bias, we conducted a structured quality appraisal tailored to the design of each included study. This review adopted established critical appraisal tools to assess the methodological quality of qualitative, quantitative, and mixed-methods studies (Crawford et al., 2024; Stern et al., 2020). For qualitative studies, we employed the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Qualitative Research, focusing on congruence between research questions, methodology, data collection and analysis, reflexivity, representation of participants' voices, and ethical considerations (Evans et al., 2025). For quantitative studies, we applied JBI checklists appropriate to each study design, assessing elements such as sample selection, validity of measurement tools, statistical analysis, and control for confounders (Barker et al., 2023). Mixed-methods studies were appraised using the Mixed Methods Appraisal Tool (MMAT, 2018 version), which evaluates the methodological quality of both the qualitative and quantitative strands, the rationale for combining methods, and the integration and interpretation of findings across strands (Hong et al., 2023; Stern et al., 2020).

Each study was independently evaluated by two reviewers. Discrepancies in ratings were discussed and resolved by consensus. When consensus could not be reached, a third reviewer was consulted. Studies were classified as high, moderate, or low in methodological quality based on tool-specific guidance and the presence of major limitations affecting internal validity or credibility of findings.

Overall, most included studies demonstrated high methodological quality, supporting the reliability of evidence on how generative AI and large language models (LLMs) impact critical thinking in EFL contexts (see Table 3). Qualitative studies, assessed using the JBI Critical Appraisal Checklist, showed strong congruence between research questions, methodologies, and data analysis. Clear researcher reflexivity, ethical considerations, and rich representation of participants' voices were common strengths, though minor concerns emerged regarding sampling scope or incomplete ethical reporting (Hapsari & Wu, 2022; Yilin et al., 2023). Quantitative studies evaluated via the JBI-Quanti checklist were generally of moderate to high quality. However, sampling bias



**Fig. 1** PRISMA flow diagram of the systematic literature review

and limited depth of statistical analysis were noted (Darmawansah et al., 2024; Lee & Davis, 2024), potentially affecting the generalizability of findings on learners' cognitive development. Mixed-methods studies, appraised using the MMAT, also demonstrated high overall quality, with clear justification for methodological integration and adequate alignment between qualitative and quantitative components. Nonetheless, a few studies showed inconsistencies in interpreting divergent findings or faced challenges related to technological implementation and participant readiness in AI-enhanced learning environments (Liu & Wang, 2024; Xhaferi & Musa, 2024).

#### Literature coding and data input

To systematically analyze the selected articles, a comprehensive coding scheme was developed to extract and categorize key data points relevant to the research objectives. The coding framework included 7 main dimensions: (1) bibliographic information including author, year, country, sample size, target population, age range, and subject; (2) research methodology including qualitative, quantitative and mixed methods; (3)

**Table 3** Quality assessment of the included qualitative, quantitative and mixed-method studies

| Study                      | Method type        | Appraisal tool | Quality rating | Notes                                                                                                                                                                                                                                          |
|----------------------------|--------------------|----------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Avsheniuk et al. (2024)    | Mixed-methods      | MMAT           | High           | Clear rationale for using mixed methods; Strong integration of qualitative and quantitative components; Adequate handling of divergences between methods; Ethical considerations addressed; Comprehensive analysis and interpretation          |
| Aymen and Zakarya (2024)   | Mixed-methods      | MMAT           | Moderate       | Clear rationale for using mixed methods; Adequate integration of qualitative and quantitative components; Some inconsistencies between qualitative and quantitative results need further clarification                                         |
| Darmawansah et al., (2024) | Quantitative study | JBIR-Quanti    | High           | Potential limitations in randomization                                                                                                                                                                                                         |
| Datskiv et al. (2024)      | Qualitative study  | JBIR-Qual      | High           | Clear reflexivity and congruence; Strong representation of participants' voices; Ethical considerations addressed; Adequate integration of methodology and research objectives                                                                 |
| Hading et al. (2024)       | Qualitative study  | JBIR-Qual      | High           | Clear reflexivity and congruence; Strong representation of participants' voices; Adequate ethical considerations                                                                                                                               |
| Hapsari and Wu (2022)      | Qualitative study  | JBIR-Qual      | Moderate       | Clear reflexivity and congruence but ethical approval not mentioned                                                                                                                                                                            |
| Khan (2024)                | Qualitative study  | JBIR-Qual      | High           | Clear reflexivity and congruence; Strong alignment between methodology and research questions; Adequate representation of participants' voices                                                                                                 |
| Lee and Davis (2024)       | Quantitative study | JBIR-Quanti    | Moderate       | Sampling bias concerns and limited statistical analysis                                                                                                                                                                                        |
| Liu and Wang (2024)        | Mixed-methods      | MMAT           | High           | Clear rationale for mixed-methods approach; Effective integration of qualitative and quantitative components; Adequate interpretation of combined results; Minor concerns regarding initial hesitation in using AI tools and need for training |
| Nabilla et al. (2024)      | Qualitative study  | JBIR-Qual      | High           | Clear reflexivity and congruence; Strong integration of qualitative methods; Adequate representation of participants' voices                                                                                                                   |
| Octavio et al. (2024)      | Qualitative study  | JBIR-Qual      | High           | Clear reflexivity and congruence; Strong alignment between methodology and research questions; Adequate representation of participant's voice                                                                                                  |

**Table 3** (continued)

| Study                   | Method type       | Appraisal tool | Quality rating | Notes                                                                                                                                                                                                                    |
|-------------------------|-------------------|----------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rizkiani et al. (2024)  | Qualitative study | JB1-Qual       | High           | Clear reflexivity and congruence; Strong integration of qualitative components; Ethical considerations addressed                                                                                                         |
| Xhaferi and Musa (2024) | Mixed-methods     | MMAT           | Moderate       | Clear rationale for mixed methods; Effective integration of qualitative & quantitative method; Adequate interpretation of combined results; Some concerns regarding sampling bias and technological challenges           |
| Yilin et al. (2023)     | Qualitative study | JB1-Qual       | High           | Clear reflexivity and congruence; Strong representation of participants and their voices; Ethical considerations addressed; Minor concerns regarding sampling size and scope                                             |
| Zhang et al. (2024)     | Mixed-methods     | MMAT           | High           | Clear rationale for mixed methods; Strong integration of qualitative and quantitative components; Adequate interpretation of combined results; Some concerns regarding uniformity of critical thinking skill development |

outcomes and findings including positive and negative impacts on CT; (4) main focuses about generative-AI tool specifications and LLMs on CT; (5) theoretical underpinnings; (6) limitations and challenges; and (7) future directions. Each article was carefully read and coded using this scheme.

For the purpose of clarity and conciseness, only five key categories: Authors (Year), Regions, Study Design, Main Focuses, and Limitations, were presented in Appendix 1. Other coded dimensions, such as theoretical frameworks and detailed outcome patterns, were retained for internal synthesis and interpretation but not displayed in the appendix.

The coding results were then transformed into binary data upon agreement between the two raters. The result of the kappa agreement test is 0.96 (96.0%; Table 4), which is considered acceptable and a high-level agreement. However, the coding result should be totally in agreement. Thus, the coding results were rechecked. The finding indicated disagreements in terms of the theory's category. It is: one rater regarded Reinforcement Learning from Human Feedback (RLHF) as cognitive theories. After a consensus, Reinforcement Learning from Human Feedback (RLHF) should be categorized under AI-based learning models. Finally, it rendered the agreement 100%. After coding, the categorized data were input into Excel to facilitate comparative analysis and thematic synthesis.

## Results

### Do generative AI tools and LLMs kill or nurture critical thinking in EFL education?

The impact of Generative Artificial Intelligence (AI) tools and Large Language Models (LLMs) on learners' critical thinking (CT) in English as a Foreign Language (EFL) education

is multifaceted, presenting both opportunities and challenges. Based on an in-depth analysis of 15 selected relevant articles, Generative AI and LLMs has the potential to nurture CT: (1) improving argumentative speaking performance and writing quality, (2) fostering analytical thinking, reasoning, evaluation, decision-making, and problem-solving skills, (3) enhancing creativity and collaboration, (4) increasing engagement, motivation, and confidence, and (5) supporting self-directed and personalized learning. However, there is also a risk of overreliance on Generative AI and LLMs, which may lead to passive thinking, a decline in analytical depth, impeding autonomous skill development, and reducing critical thinking and creativity.

On the positive side, based on 10 studies (66.67%), generative AI tools and LLMs-powered tools have been found to enhance students' CT in EFL education. Lee and Davis (2024) observed that AI-supported writing tasks encouraged CT, while Avsheniuk et al. (2024) noted improvements in decision-making, analytical thinking, and creativity in EFL environments. Similarly, Nabilla et al. (2024) found that Paragraph AI Technology improved argument structure, logic flow, and evidence evaluation, all of which are essential for CT development in English language courses. Furthermore, generative AI-assisted teaching methods, as explored by Darmawansah et al., (2024), significantly enhanced English students' argumentative speaking performance, CT awareness, and collaboration tendency. Datskiv et al. (2024) stated that ChatGPT can stimulate EFL learners' CT by providing access to information, supporting self-study, and promoting problem-solving skills. Additionally, Liu and Wang (2024) found that AI tools like ChatGPT and SummarizBot significantly improved EFL students' CT skills in the experimental group compared to the control group. Supporting this, Xhaferi and Musa (2024) highlighted that both EFL students and professors recognize the role of AI tools, such as ChatGPT 4.0, in enhancing CT, improving language expertise, increasing motivation and engagement, identifying problems, fostering contextual and cultural understanding, personalizing learning, and strengthening problem-solving skills in EFL learning classes. Moreover, Zhang et al. (2024) demonstrated that 71.93% of revised drafts reflected modifications in claims, warrants, and grounds in a EFL classroom, indicating the application of CT skills. Their study also categorized generative AI-student interactions into three themes: making suggestions for language revisions, extending and developing main points, and providing supporting materials for arguments (Zhang et al., 2024). Furthermore, Hapsari and Wu (2022) and Octavio et al. (2024) reported that teachers believed AI chatbots learning model benefit the learning process and cognitive skills, and contribute to achieving course objectives and improving user's confidence, so that it can foster EFL learners' CT.

However, 5 studies (33.33%) caution against the drawbacks of generative AI and LLMs tools. Khan (2024) reported that students displayed a decline in analytical depth after using generative AI, relying more on AI-generated content rather than engaging in deep critical analysis. Yilin et al. (2023) and Hading et al. (2024) also emphasized the risk of student overreliance on generative AI in EFL education, which could impede skill development.

**Table 4** Kappa agreement test for coding

| Agreement | Kappa | Std. Err | T <sup>b</sup> | Prob Sig |
|-----------|-------|----------|----------------|----------|
| 96.0%     | 0.96  | 0.028    | 15.497         | < 0.001  |



Nevertheless, they acknowledged generative AI's role in assisting information retrieval, data analysis, problem-solving, quick information evaluation, exploring alternative ideas, and improving reasoning abilities. Additionally, Aymen and Zakarya (2024) pointed out that while generative AI can expose students to diverse perspectives, it may also lead to passive thinking and reduced creativity if students become overly dependent on AI-generated responses in EFL courses. Rizkiani et al. (2024) found that although AI-generative tools significantly improved EFL students' abilities to create cohesive arguments, fix language, and incorporate evidence, overreliance on generative AI may impede the development of autonomous analytical skills and CT skills.

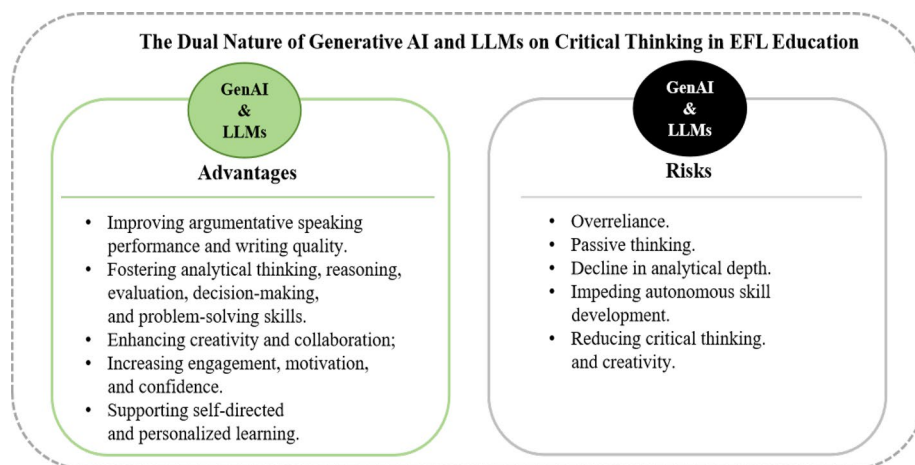
To maximize the benefits of AI-driven generative tools while mitigating potential drawbacks, educators must implement strategies that foster critical engagement in EFL classroom. Octavio et al. (2024) stressed the importance of AI literacy, including prompt crafting and critical evaluation of AI outputs, to ensure reliability and minimize biases. Yilin et al. (2023) recommended incorporating group discussions, project-based learning, and writing exercises to encourage independent thinking and CT in EFL teaching process.

Overall, as shown in Fig. 2, generative AI and LLMs-powered tools can both nurture and hinder CT in EFL teaching and learning. Their effectiveness depends on how they are integrated into EFL educational practices. When used thoughtfully, with an emphasis on student agency, critical evaluation, and interactive learning, generative AI and LLMs can be a powerful tool to develop and enhance CT skills. However, unchecked reliance on AI-generated content may lead to passive learning and hinder deep analytical engagement. Therefore, a balanced and guided approach is essential for leveraging generative AI and LLMs' potential in fostering CT in EFL context.

#### **What research methods were employed in generative AI Tools and LLMs for fostering critical thinking in EFL education?**

The research methods used in these studies predominantly consist of qualitative and mixed-method approaches, with a smaller representation of quantitative research in Fig. 3.

Firstly, 55.33% ( $n=8$ ) of publications are published using qualitative research methods, focusing on in-depth exploration and analysis of Generative AI tools and Large Language Models (LLMs)-powered tools in critical thinking (CT) development within English as a Foreign Language (EFL) education context. These studies employed methods including interviews, case studies, and thematic analysis to provide rich, descriptive insights. Case studies, such as those conducted by Octavio et al. (2024) and Rizkiani et al. (2024), examined specific contexts in depth, offering rich, contextualized findings. Furthermore, qualitative studies incorporating interviews were widely used. Yilin et al. (2023) conducted semi-structured interviews, while Dat-skiv et al. (2024) employed focus group discussions with thematic analysis to identify key patterns. Other researchers, Hading et al. (2024) and Nabilla et al. (2024), applied descriptive qualitative approaches, providing detailed narratives based on direct interviews. Additionally, Khan (2024) performed a qualitative content analysis, comparing student assignments from different periods and supplementing the analysis with teacher interviews. Hapsari and Wu (2022) adopted a qualitative research



**Fig. 2** The dual nature of generative AI and LLMs on CT in EFL education

approach, utilizing in-depth interviews to investigate how AI chatbots function as self-regulated learning tools.

Additionally, 33.33% ( $n = 5$ ) of the analyzed papers employed mixed methods. These studies integrated both qualitative and quantitative research techniques to achieve a more comprehensive understanding of their research questions on generative AI and LLMs in CT development in EFL classroom. Zhang et al. (2024) conducted a study incorporating both survey data and qualitative reflections, while Avsheniuk et al. (2024) and Xhaferi and Musa (2024) also adopted mixed-methods approaches to provide a balanced analysis. Similarly, Aymen and Zakarya (2024) employed questionnaires alongside interviews to collect both statistical and interpretive data, while Liu and Wang (2024) designed an eight-week intervention study employing mixed methods. This approach allows for triangulation, enhancing the validity and reliability of findings by combining qualitative insights with statistical analysis.

Beyond qualitative approaches and mixed methods, there are two quantitative studies. Experimental and intervention-based studies played a crucial role in evaluating the effects of AI-generative tools and LLMs on CT in EFL education. Darmawansah et al., (2024) conducted a quasi-experimental study to measure changes over time, and Lee and Davis (2024) adopted a quantitative research methodology to collect data.

#### **What were the main research focuses of generative AI Tools and LLMs for fostering critical thinking in EFL education?**

Through a careful review of 15 studies, research on generative Artificial Intelligence (AI) tools and Large Language Models (LLMs) in critical thinking (CT) development within English as a Foreign Language (EFL) education revolves around three key themes: (1) the impact of generative AI and LLMs on EFL students' CT skills, (2) generative AI's role in EFL teaching and learning, and (3) the influence of generative AI on EFL teachers' competencies and instructional practices.

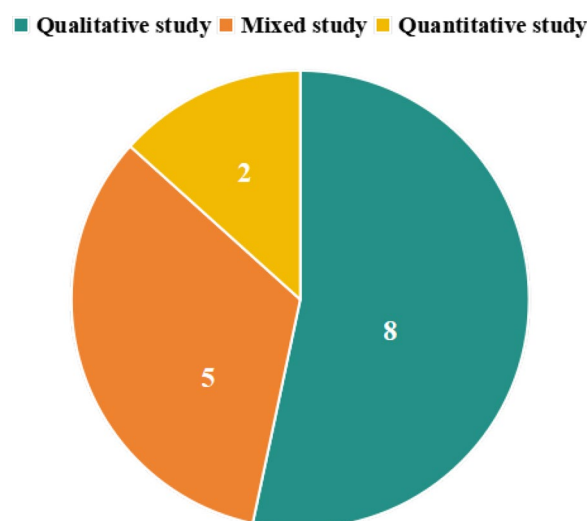
Firstly, eight studies examined how generative AI and LLMs tools on EFL students' CT skills. Datskiv et al. (2024) highlighted ChatGPT's potential to enhance self-study,

creativity, and problem-solving skills, while Yilin et al. (2023) explored its role in actively fostering CT in EFL classrooms. Liu and Wang (2024) further examined how AI tools like ChatGPT and SummarizBot contributed to the development of CT skills in English literature classes in China. Similarly, Xhaferi and Musa (2024) explored both the potential benefits and challenges of integrating ChatGPT 4.0 in English language learning.

Conversely, other researchers cautioned against over-reliance on generative AI and LLMs-powered tools, arguing that it might hinder creativity and intellectual independence (Aymen & Zakarya, 2024; Hading et al., 2024; Khan, 2024; Rizkiani et al., 2024; Yilin et al., 2023). Similarly, Hading et al. (2024) investigated both the positive and negative perceptions of generative AI's impact on EFL students' CT development. Lee and Davis (2024) proposed that generative AI in writing tasks encouraged students to interact with the AI to produce well-structured paragraphs, reinforcing CT skills.

Furthermore, five studies explored generative AI's role in EFL teaching and learning. Rizkiani et al. (2024) and Avsheniuk et al. (2024) examined how AI-powered tools, such as ChatGPT and Paragraph AI, support students in developing CT through structured writing exercises. Darmawansah et al., (2024) analyzed how ChatGPT fosters argumentation skills by guiding students through collaborative learning stages, reinforcing the idea that generative AI can enhance structured reasoning. Hapsari and Wu (2022) investigated AI chatbots' impact on English speaking skills, emphasizing their ability to alleviate speaking anxiety, boost enjoyment, and foster CT among EFL learners. Zhang et al. (2024) further explored how AI chatbots improved university EFL students' speech script writing quality while simultaneously enhancing their CT skills.

Beyond student learning, three studies addressed the role of generative AI on teachers' competencies and instructional practices. Octavio et al. (2024) emphasized the importance of EFL teachers developing CT skills alongside AI competencies to effectively assess AI-generated content. Nabilla et al. (2024) investigated the broader implications of AI tools like ChatGPT in language learning, examining how they shape both student and teacher engagement with CT. Additionally, Xhaferi and Musa (2024) discussed the



**Fig. 3** The distribution of research methods

challenges of ChatGPT 4.0 integration poses for EFL instructors in fostering students' CT skills.

### **What were theory or models involved in generative AI Tools and LLMs for fostering critical thinking in EFL education?**

The selected 15 studies of generative artificial intelligence (AI) tools and large language models (LLMs) in critical thinking (CT) development within English as a Foreign Language (EFL) education is supported by various theoretical and conceptual frameworks. These frameworks can be categorized into four main groups: cognitive theories, sociocultural theories, critical thinking and argumentation frameworks, and AI-based learning models. Each of these perspectives highlights different ways in which generative AI can influence and enhance CT in EFL teaching and learning.

Firstly, cognitive theories focus on how learners process, store, and retrieve information. Xhaferi and Musa (2024) applied Bloom's Taxonomy, which categorized cognitive processes from basic recall to higher order thinking such as evaluation and creation. Similarly, Lee and Davis (2024) explored Constructivist Learning Theories, emphasizing knowledge construction through active engagement and interaction. Likewise, Khan (2024) examined Emergentism and Mentalism, which offered different perspectives on cognitive development. These cognitive perspectives suggest that AI-generative tools can scaffold thinking processes and facilitate deeper, structured reasoning.

Furthermore, sociocultural theories highlight the social nature of learning and the role of interaction in cognitive development. Khan (2024) integrated Vygotsky's Sociocultural Theory, which emphasized the importance of social interaction and scaffolding in knowledge acquisition. Similarly, Liu and Wang (2024) focused on the Zone of Proximal Development (ZPD), which underscored the potential of generative AI and LLMs as a learning mediator, providing tailored guidance to help learners progress. Furthermore, Yilin et al. (2023) adopted Interaction Theory, which emphasized EFL learning through communicative exchanges and interactive discussions. Additionally, Darmawansah et al., (2024) utilized Collaboration Scripts to structure collaborative learning in EFL tasks. These perspectives suggest that Generative AI-driven learning environments foster collaboration, a key element in CT development.

Additionally, critical thinking and argumentation frameworks provide structured models for logical reasoning and analytical skills. Avsheniuk et al. (2024) and Nabilla et al. (2024) applied Facione's Theoretical Framework on Critical Thinking, which defined key cognitive skills such as analysis, inference, and evaluation. Xhaferi and Musa (2024) incorporated Paul and Elder's Critical Thinking Framework, which outlined essential intellectual standards and elements of reasoning. Aymen and Zakarya (2024) integrated multiple frameworks, including the Paul-Elder Framework, the Ennis Model, and Argumentation Models, emphasizing logical reasoning and structured argumentation. Additionally, Darmawansah et al., (2024) employed Argument Mapping to enhance the organization of reasoning and argumentation in critical thinking tasks. These models align well with generative AI's ability to foster CT skills.

Finally, AI-based learning models explore how artificial intelligence supports personalized and adaptive learning. Nabilla et al. (2024) and Avsheniuk et al. (2024) considered UNESCO's Definition of AI, which provided an ethical and educational framework

for AI-driven learning environments. Zhang et al. (2024) applied the Production-Oriented Approach, emphasizing knowledge construction through output-driven learning. Another approach, discussed by Yilin et al. (2023), involved Reinforcement Learning from Human Feedback (RLHF), an AI-driven model where learning was refined based on user interactions. These AI-based models highlight the adaptive nature of generative AI, allowing for customized learning experiences that enhance EFL students' CT abilities.

Overall, the integration of generative AI tools and LLMs in CT within EFL education is underpinned by four categories of theories and models based on Fig. 4. Cognitive theories emphasize structured knowledge construction, sociocultural theories stress the importance of interaction, critical thinking and argumentation frameworks provide structured critical thinking models, and AI-based learning models showcase technology's potential for personalized learning. Together, these perspectives offer a comprehensive understanding of how generative AI and LLMs-powered tools can support and enhance CT in EFL education.

#### **What were the limitations or challenges identified in generative AI Tools and LLMs for fostering critical thinking in EFL education?**

Obviously, the integration of generative artificial intelligence (AI) tools and large language models (LLMs) in fostering critical thinking (CT) within English as a Foreign Language (EFL) education has shown potential benefits, yet four limitations and challenges hinder their effectiveness in CT skills in EFL courses. These challenges can be categorized into issues related to research scope, user dependency, generative AI reliability, and pedagogical integration.

Firstly, eight studies have limitations in research scope and generalizability. Research conducted by Lee and Davis (2024) was constrained by a short timeframe, making it difficult to assess the long-term effects of generative AI on EFL learning. Similarly, studies by Octavio et al. (2024), Liu and Wang (2024), and Khan (2024) were restricted to specific cases, institutions, and geographic regions, reducing the applicability of findings to broader educational contexts. Small sample sizes, as noted in the research of Avsheniuk et al. (2024), Liu and Wang (2024), Hapsari and Wu (2022), and Aymen and Zakarya (2024), further limited the robustness of conclusions. Additionally, Xhaferi and Musa (2024) noted that the research topic itself was relatively new, which might have affected the depth and breadth of the study.

Secondly, there is a risk of overreliance on AI-generative technology, potentially hindering EFL students' development of independent CT skills. Researchers warned that excessive dependence on AI-generated content could reduce students' ability to analyze and evaluate information autonomously (Aymen & Zakarya, 2024; Hading et al., 2024; Khan, 2024; Rizkiani et al., 2024). Similarly, Nabilla et al. (2024) emphasized that generative AI should be a supplementary tool rather than a replacement for EFL students' cognitive efforts. Liu and Wang (2024) further raised concerns about the reliance on self-reported measures of CT, which may not always provide an accurate assessment of EFL students' actual progress.

Thirdly, generative AI tools and LLMs themselves present notable limitations in reliability. Datskiv et al. (2024) highlighted that ChatGPT struggled with content reliability,

sometimes generating misinformation and failing to differentiate between credible and non-credible sources. Hading et al. (2024) also pointed out the rapid technological evolution of AI-driven generative tools, making it challenging for EFL educators and learners to keep pace with its changes. Additionally, the quality of AI-generated responses heavily depended on the quality of user prompts, as noted by Darmawansah et al., (2024).

Lastly, pedagogical challenges remain, as educators are still in the early stages of integrating generative AI and LLMs-powered tools into EFL teaching practices. Yilin et al. (2023) found that teachers were still exploring the use of ChatGPT, which might affect the effectiveness of its implementation in developing CT. Furthermore, Octavio et al. (2024) noted that limitations in ChatGPT (free vs. premium models) could impact the quality of educational outcomes. Additionally, Zhang et al. (2024) found that 28% of EFL participants failed to exhibit progress in CT skills, which might be linked to the lack of targeted guidance on leveraging the AI chatbot effectively.

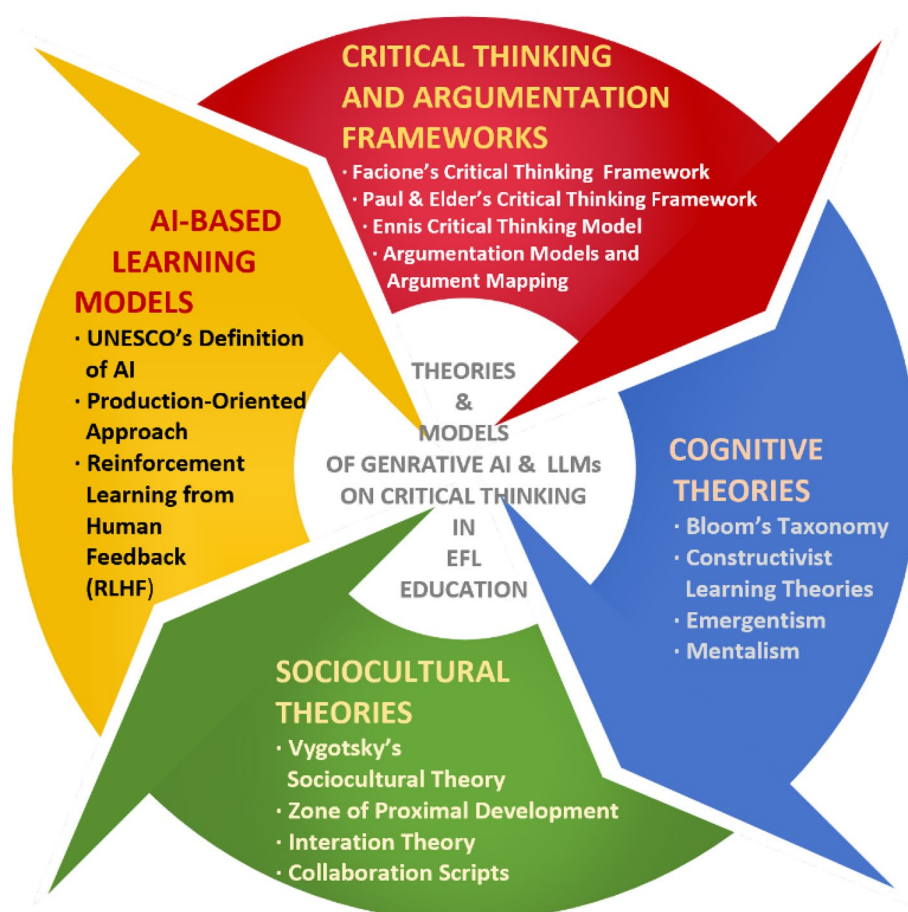
#### **What are the future directions for generative AI Tools and LLMs for fostering critical thinking in EFL education?**

As generative Artificial Intelligence (AI) tools and Large Language Models (LLMs) within continue to be integrated into fostering critical thinking (CT) within English as a Foreign Language (EFL) education, future research must address key areas to maximize their effectiveness in EFL learners' CT development. These future directions can be categorized into four categories: long-term impact studies, broader research scope, responsible AI integration, and pedagogical improvements.

Firstly, six studies emphasize the need for long-term research on generative AI's impact on EFL learning and CT. Lee and Davis (2024) suggested examining the sustained effects of AI tools in diverse educational contexts, while Khan (2024) and Rizkiani et al. (2024) highlighted the importance of studying AI's influence on cognitive development and academic performance over extended periods. Nabilla et al. (2024) also recommended longitudinal studies to assess the long-term impact of AI-assisted writing tools on EFL students' analytical and writing skills. Similarly, Hapsari and Wu (2022) stressed the necessity of conducting longitudinal studies to evaluate how AI-generated content influences EFL students' critical thinking over time, and Xhaferi and Musa (2024) noted that further research should examine ChatGPT 4.0's impact on various age groups and collect feedback from EFL educators who have implemented it to gain a more comprehensive understanding of its educational implications.

Secondly, expanding the scope of research is crucial to understanding generative AI's effects across different demographics and educational settings. Yilin et al. (2023), Liu and Wang (2024), and Avsheniuk et al. (2024), Hapsari and Wu (2022) proposed increasing sample sizes and incorporating diverse geographical and cultural contexts. Octavio et al. (2024) further suggested studying the effectiveness of ChatGPT across different age groups, ensuring that findings were applicable to a wide range of learners. Liu and Wang (2024) also recommended exploring various educational contexts and examining the combined impact of generative AI tools and pedagogical approaches to develop more comprehensive strategies for generative AI integration.





**Fig. 4** Theories and models of generative AI and LLMs on CT in EFL Education

Thirdly, responsible generative AI and LLMs-powered tools integration remains a key focus. Datskiv et al. (2024) called for research on generative AI's role in EFL teacher education and the development of guidelines for ethical and effective generative AI use. Similarly, Nabilla et al. (2024) and Hading et al. (2024) emphasized the need for structured policies that guided students in using generative AI tools responsibly, preventing overreliance while promoting independent CT skills. Xhaferi and Musa (2024) recommended that EFL teachers incorporated ChatGPT 4.0 into their instructional strategies while continuously refining integration methods and preparing for potential technical challenges.

Finally, improving generative AI-assisted pedagogical strategies can enhance learning outcomes. Darmawansah et al., (2024) recommended integrating ChatGPT with established teaching methods and developing personalized learning paths tailored to EFL students' proficiency levels. Lee and Davis (2024) also highlighted the importance of embedding generative AI into language learning curriculums in a way that complemented traditional teaching approaches. Zhang et al. (2024) suggested that educators devised more precise and pedagogically aligned conversational strategies with generative AI, customizing them to specific educational objectives, content areas, and EFL students' learning habits.



## Discussion

The rapid development of generative Artificial Intelligence (AI) technology tools and large language models (LLMs) is profoundly impacting the field of English as a Foreign Language (EFL) education, driving continuous innovation in English educational concepts and teaching methods. Among these, critical thinking (CT), as a key skill, has garnered increasing attention for its application in EFL education (Kennedy & Sundberg, 2020). From 2022 to 2025, research on AI-generative tools and LLMs-powered for CT in the context of EFL education has shown significant growth. This paper provides an in-depth review of the impact of generative AI tools and LLMs on CT within EFL education context based on 15 selected articles. Key research questions for this study are discussed below.

### The dual nature of generative AI tools and LLMs on critical thinking in EFL education

The findings from the reviewed studies collectively highlight both the advantages and the risks of AI-generative tools and LLMs in fostering CT in EFL teaching and learning. While ten studies (Avsheniuk et al., 2024; Darmawansah et al., 2024; Datskiv et al., 2024; Hapsari & Wu, 2022; Lee & Davis, 2024; Liu & Wang, 2024; Nabilla et al., 2024; Octavio et al., 2024; Xhaferi & Musa, 2024; Zhang et al., 2024) provided evidence that generative AI tools and LLMs could enhance EFL learners' CT, mainly through improving argumentative speaking performance and writing quality, fostering analytical thinking, reasoning, evaluation, decision-making, and problem-solving skills, enhancing creativity and collaboration, increasing engagement, motivation, and confidence, and supporting self-directed and personalized learning, others (Aymen & Zakarya, 2024; Hading et al., 2024; Khan, 2024; Rizkiani et al., 2024; Yilin et al., 2023) raised concerns about EFL learners' overreliance on generative AI and LLMs, which led to passive engagement, resulting in passive thinking, declining analytical depth, impeding autonomous skill development, and reducing critical thinking and creativity. This duality suggests that generative AI tools and LLMs' impact is largely dependent on how it is integrated into educational practices and the extent to which students engage critically with AI-generated content.

A key reason for the positive effects of AI-generative tools may lie in their ability to scaffold learning. Studies have shown that generative AI facilitates structured guidance in writing, speaking, and reasoning (Darmawansah et al., 2024; Datskiv et al., 2024), enabling students to refine their ideas, evaluate arguments, and construct coherent responses in EFL classrooms. From a cognitive perspective, Vygotsky's (1978) theory of the Zone of Proximal Development (ZPD) might explain this benefit, as generative AI functions as a dynamic support system, offering prompts, feedback, and corrective suggestions that help learners operate beyond their current level of competence (Khan, 2024; Liu & Wang, 2024). Moreover, the interactive nature of AI chatbots, as reported by Yilin et al. (2023) and Zhang et al. (2024), suggested that generative AI-assisted revisions encouraged EFL students to refine claims and justify arguments, reinforcing metacognitive awareness, which was a crucial element of CT.

Despite these advantages, concerns regarding AI-induced passivity highlight a critical challenge. The tendency for students to rely on AI-generated content without deep engagement (Khan, 2024; Rizkiani et al., 2024) may be explained by cognitive load

theory (Skulmowski & Xu, 2022). When AI- generative tools simplify complex tasks, EFL students might bypass the effortful cognitive processes necessary for deep learning. Additionally, social constructivist perspectives (Hading et al., 2024; Lee & Davis, 2024; Yilin et al., 2023) emphasize the importance of interaction in CT development, which AI tools, when used in isolation, may not fully replicate. While Generative AI provides access to diverse perspectives (Aymen & Zakarya, 2024), there is a risk that EFL students engage superficially with such insights rather than critically evaluating them.

Based on challenges faced, AI literacy has been identified as a crucial factor (Octavio et al., 2024). Developing EFL students' ability to critically assess AI-generated content, question its validity, and integrate it into broader discussions can mitigate risks of passivity. Furthermore, instructional strategies such as collaborative learning (Darmawansah et al., 2024) and inquiry-based approaches (Yilin et al., 2023) have been suggested to complement AI-generative tools, ensuring that EFL students remain active participants in the learning process.

Overall, the findings suggest that while generative AI has the potential to enhance CT, its effectiveness depends on structured implementation. Rather than replacing critical engagement, generative AI tools and LLMs-powered tools should serve as a support that prompts deeper inquiry and reflection. Future research could explore optimal instructional designs that balance generative AI integration with active learning strategies, ensuring that EFL learners develop both technological competence and independent analytical skills.

#### **The research methods for generative AI Tools and LLMs on critical thinking in EFL education**

Among the analyzed articles, a most portion utilized qualitative research methods, including case studies and interviews, and focus group to explore AI-driven generative tools and LLMs on CT in EFL education context (Datskiv et al., 2024; Hading et al., 2024; Hapsari & Wu, 2022; Khan, 2024; Nabilla et al., 2024; Octavio et al., 2024; Rizkiani et al., 2024; Yilin et al., 2023). Additionally, five studies adopted mixed-methods designs, combining qualitative and quantitative techniques to provide a more comprehensive understanding of generative AI tools and LLMs on CT within EFL teaching and learning (Avsheniuk et al., 2024; Aymen & Zakarya, 2024; Liu & Wang, 2024; Xhaferi & Musa, 2024; Zhang et al., 2024). Quantitative approach was also employed in two studies, involving pre- and post-tests as well as questionnaire surveys to measure CT outcomes in the context of utilizing LLMs-driven generative AI tools in EFL courses (Darmawansah et al., 2024).

These findings are largely consistent with Melisa et al. (2025), who observed a similar dominance of qualitative and mixed approaches in technology-enhanced CT studies. However, in contrast to earlier systematic reviews like Sardi et al. (2025), which noted the scarcity of rigorous mixed-methods studies, and this review found that mixed designs are gaining traction, particularly in recent post-ChatGPT publications. This shift may reflect a growing recognition of the need to triangulate learner experience, performance data, and AI interaction processes, especially as LLMs play an increasingly dialogic and embedded role in the EFL classroom. The increasing complexity of AI-enhanced learning environments seems to necessitate richer, more integrative methodological

frameworks capable of capturing not only outcomes but also the underlying reasoning, engagement, and co-construction dynamics.

### **The main research focuses of generative AI Tools and LLMs on critical thinking in EFL education**

The thematic analysis of the 15 reviewed studies revealed three major focuses concerning how generative AI tools and large language models (LLMs) influence critical thinking (CT) in EFL education: their effects on students' CT skills, their integration in language learning tasks, and their implications for EFL teachers' roles.

Firstly, consistent with recent findings (Datskiv et al., 2024; Liu & Wang, 2024; Xhaferi & Musa, 2024; Yilin et al., 2023), many studies emphasized that generative AI can enhance learners' CT by enabling independent exploration, facilitating argument construction, and supporting information synthesis. These findings contrast with earlier literature, such as Fitria (2023), which regarded AI tools primarily as assistive devices for translation or factual retrieval, with minimal emphasis on higher-order thinking. This shift may be attributed to the advent of more advanced LLMs like ChatGPT, which enable dialogic interaction and more complex task completion. However, echoing the concerns of Aymen and Zakarya (2024), Hapsari and Wu (2022) and Khan (2024), several studies warned against over-reliance on AI, which may lead to reduced learner autonomy and superficial engagement. This duality implies that AI's contribution to CT development depends not on the technology itself, but on pedagogical framing and critical scaffolding.

Additionally, the reviewed literature increasingly focuses on the application of generative AI in EFL writing and language learning processes (Avsheniuk et al., 2024; Darmawansah et al., 2024; Rizkiani et al., 2024; Zhang et al., 2024). Generative tools were widely used to help learners improve textual coherence, grammar accuracy, and organization. While this aligns with Lee and Davis (2024), who also highlighted the benefits of AI-assisted language production, a new insight from this review is the concern that AI support in writing tasks may not always activate critical reasoning. If used primarily for surface-level correction rather than for idea development or evaluation, its influence on CT might remain limited. This highlights the need for instructional design that embeds reflective and analytical tasks around AI-generated outputs.

Third, several studies expanded the discussion to EFL teachers' professional development, noting that generative AI also reshapes instructional roles and classroom interaction (Nabilla et al., 2024; Octavio et al., 2024; Xhaferi & Musa, 2024). Rather than acting as mere information providers, teachers are increasingly expected to orchestrate AI-mediated learning experiences, guiding students to critique and refine AI-generated content. This aligns with broader shifts noted by Nurhidayat et al. (2024), who argue that digital literacy and CT facilitation are becoming central components of teacher competency frameworks in EFL settings.

Overall, the evolving focus from tool utility to cognitive and pedagogical integration marks a significant progression in the field, with this review capturing nuanced understandings of both the affordances and limitations of generative AI in supporting CT development.

### Theory and models in generative AI Tools and LLMs on critical thinking in EFL education

The findings highlight how generative AI tools and LLMs intersect with multiple theoretical frameworks to support CT in EFL education. While cognitive theories, sociocultural theories, critical thinking and argumentation frameworks, and AI-based learning models provide distinct perspectives, they collectively suggest that generative AI can act as both a scaffold for CT and a facilitator of interactive learning environments. However, the extent to which generative AI enhances CT depends on how these frameworks are integrated into EFL educational practice.

From a cognitive perspective, generative AI aligns with Bloom's Taxonomy (Xhaferi & Musa, 2024) and constructivist approaches (Lee & Davis, 2024) by fostering progressively complex cognitive processes. The ability of LLMs to guide EFL learners from basic comprehension to higher-order thinking suggests a strong theoretical foundation for AI-supported cognitive development. Emergentism and Mentalism (Khan, 2024) in generative AI research, which provide further insight into cognitive development. Emergentism suggests that learning emerges from complex interactions between environmental input and cognitive structures, reinforcing the adaptability of generative AI tools in personalized learning environments. In contrast, Mentalism emphasizes the role of innate cognitive mechanisms in language acquisition and CT, highlighting potential constraints in generative AI-driven learning, particularly regarding implicit reasoning processes. There is a potential limitation, the overreliance on AI-generated prompts, which may reduce EFL students' ability to independently construct arguments without external guidance. This raises questions about how to balance generative AI's scaffolding role with the need for EFL learners to develop autonomy in CT.

Sociocultural theories emphasize the social and contextual dimensions of learning, reinforcing the idea that generative AI should not function as a solitary tutor but as a mediator of collaborative learning. The Zone of Proximal Development (ZPD) (Liu & Wang, 2024) and Vygotsky's Sociocultural Theory (Khan, 2024) suggested that learning occurs most effectively through guided interaction. Generative AI-powered tools that facilitate peer discussion, argumentation, and structured debate align well with this framework. For instance, collaboration scripts proposed by Darmawansah et al., (2024) demonstrated how generative AI can support structured dialogue, leading to deeper engagement with multiple perspectives. However, the challenge remains in ensuring that generative AI-driven discussions retain human-like adaptability, as rigid algorithmic responses may fail to capture the complexity of human argumentation. Future research could examine how adaptive generative AI models can better simulate dynamic, real-time dialogue in collaborative learning within the EFL education.

The application of critical thinking and argumentation frameworks further supports generative AI's role in structured reasoning, CT and argumentation. The Paul-Elder Critical Thinking Framework (Xhaferi & Musa, 2024) and Facione's Critical Thinking Model (Avsheniuk et al., 2024) provide essential criteria for evaluating the logical coherence and depth of AI-generated responses. These frameworks align with generative AI and LLMs' ability to provide instant analytical feedback, helping learners refine their reasoning processes. However, while generative AI and LLMs excel at recognizing logical fallacies and structuring arguments, it lacks true metacognition, which is the ability to reflect on its own reasoning and biases. This limitation suggests that generative AI

and LLMs should be seen as a complementary tool rather than a replacement for EFL learners' CT instruction. Further empirical studies could investigate whether generative AI-augmented feedback leads to long-term improvements in EFL students' independent CT skills or whether EFL learners become passive recipients of AI-driven insights.

AI-based learning models offer another dimension, particularly in adaptive and AI output-driven learning. The Production-Oriented Approach (Zhang et al., 2024) underscores the importance of active knowledge application, which generative AI and LLMs can enhance through real-time content generation and personalized feedback loops. Meanwhile, UNESCO's Definition of AI (Avsheniuk et al., 2024) provides an ethical framework, emphasizing equitable access and responsible generative AI use in EFL education. In addition, Reinforcement Learning from Human Feedback (RLHF) (Yilin et al., 2023) further refines this process by allowing AI-generative tools to adapt based on user interactions, reinforcing personalized learning pathways. While generative AI can personalize learning experiences, concerns about data privacy, bias, and the ethical implications of automated assessment should be addressed. Future research should explore how generative AI systems can be designed to mitigate biases and ensure fairness, particularly in diverse educational settings where linguistic and cultural variations influence learning outcomes.

While this review found that most studies applied cognitive and argumentation theories as theoretical underpinnings, prior literature (Avsheniuk et al., 2024; Khaferi & Musa, 2024) reported a heavier reliance on traditional cognitive theory. This shift may reflect the recent emergence of LLM-based tools requiring more nuanced frameworks to support human-AI collaboration in thinking processes (Avsheniuk et al., 2024; Yilin et al., 2023; Zhang et al., 2024). To improve LLMs and generative AI's educational potential, future research can explore hybrid pedagogical models that integrate: Cognitive scaffolding (constructivist approaches) with adaptive AI feedback mechanisms, Dialogic learning (sociocultural perspectives) with AI-driven argumentation modeling, Ethical AI literacy frameworks to ensure responsible AI-generative tools' engagement. Moving forward, EFL educators and researchers need to establish a balanced AI-integrated pedagogy framework that combines EFL learners' CT instruction with generative AI's computational power.

### **The limitations, challenges and future directions for generative AI Tools and LLMs on critical thinking in EFL education**

The integration of generative AI tools and LLMs in EFL education has introduced both opportunities and challenges in fostering CT. While these technologies offer new ways to engage EFL students in learning, their limitations highlight concerns that need to be addressed through future research.

Firstly, a significant limitation in current studies is the narrow research scope, often constrained by small sample sizes, short study durations, limited geographic and cultural contexts. Research by Lee and Davis (2024) focused on a single semester, which might not capture generative AI's long-term impact. Similarly, studies by Avsheniuk et al. (2024), Aymen and Zakarya (2024), Hapsari and Wu (2022), Khan (2024), Liu and Wang (2024), Nabilla et al. (2024), Octavio et al. (2024), Rizkiani et al. (2024), and Khaferi and Musa (2024) were limited to specific institutions, sample size, duration, or

regions, reducing the generalizability of their findings. Future studies should adopt longitudinal research designs to track EFL students' CT development over extended periods. Additionally, cross-cultural and multi-institutional studies should be conducted to assess how AI-generative tools impacted varies across different educational systems and demographic groups. Expanding research scope provides a more comprehensive understanding of LLMs-driven generative AI's role in EFL education.

Secondly, a major concern in generative AI-assisted learning is the potential for over-reliance on AI-driven generative tools, which weaken EFL students' ability to think independently (Aymen & Zakarya, 2024; Hading et al., 2024; Khan, 2024; Liu & Wang, 2024; Nabilla et al., 2024; Rizkiani et al., 2024). If EFL students consistently turn to generative AI and LLMs-powered tools for answers rather than engaging in deep analysis, their CT skills may suffer. Future research should focus on blended learning approaches, where generative AI is integrated as a support tool rather than a replacement for traditional learning methods in EFL classrooms. Moreover, developing guidelines for responsible generative AI use can help ensure that EFL students use AI-generative technology to enhance their thinking rather than substitute it. Additionally, EFL educators should incorporate human-AI collaborative tasks where EFL students should critically evaluate generative AI responses, fostering analytical skills.

Thirdly, enhancing reliability and pedagogical adaptability of generative AI and LLMs-powered tools is important to overcoming their technical limitations. LLMs like ChatGPT were often criticized for generating misinformation, inconsistencies, and difficulties in distinguishing reliable sources (Datskiv et al., 2024). These issues raise concerns about EFL students passively accepting AI-generated responses without critically evaluating their accuracy. Additionally, Hading et al. (2024) highlighted the rapid evolution of generative AI models, making it challenging for EFL educators to keep pace with updates and integrate them effectively into curricula. Future research should focus on developing AI literacy programs that train EFL students and educators to critically assess AI-generated content. Additionally, improving prompt engineering strategies can enhance generative AI's reliability, ensuring that EFL students receive more accurate and contextually appropriate information. Research should also explore how generative AI can be fine-tuned to align with educational goals, ensuring that its responses support, rather than hinder CT development.

Additionally, teachers play a crucial role in guiding students' use of AI-generative tools, but many are still in the early stages of understanding how to integrate LLMs-driven generative AI tools effectively (Yilin et al., 2023). And Xhaferi and Musa (2024) suggested that EFL teachers integrated ChatGPT 4.0 into their teaching practices, continuously improving its implementation while anticipating possible technical difficulties. Without proper training, EFL teachers struggle to incorporate AI-generative tools in ways that truly foster CT. Future research should explore how generative AI can be integrated into EFL teacher education programs, ensuring that EFL educators are equipped with strategies to leverage generative AI and LLMs-powered tools effectively while maintaining pedagogical integrity. Additionally, studies should investigate how generative AI can be adapted to provide personalized learning experiences, catering to different proficiency levels. Generative AI's role should be complementary, which means



enhancing lesson planning, formative assessment, and feedback mechanisms without replacing the EFL teacher's role in fostering deep learning and CT.

Ultimately, while generative AI is often used to assist with EFL reading, writing and speaking courses, its potential for directly fostering CT has not been fully explored. Current research has primarily focused on generative AI's impact on language acquisition rather than its role in cognitive skill development. Future research should focus on developing AI-empowered CT frameworks, where generative AI and LLMs are used to pose thought-provoking questions, challenge assumptions, and facilitate debates. Generative AI tools should be integrated into inquiry-based learning models, where students engage in argumentative writing, evidence-based reasoning, and problem-solving tasks with generative AI acting as a cognitive partner rather than an answer provider.

Although generative AI tools and LLMs-powered tools present promising opportunities for enhancing CT in EFL education, their limitations should be carefully addressed through strategic research and pedagogical innovation. Expanding study scopes, improving AI-generative tools' reliability, preventing overreliance, enhancing teacher training, and developing generative AI-based CT frameworks in EFL education are all essential steps. Thus, striking a balance between AI-assisted learning and human-led critical inquiry, it can be ensured that generative AI and LLMs serve as tools for empowerment rather than dependency, ultimately fostering a more analytical and critically engaged generation of EFL learners.

## Conclusion

This systematic review to uncover the dual nature, research methods, main focuses, theory and models, limitations and future directions of 15 selected studies on generative Artificial Intelligence (AI) tools and Large Language Models (LLMs)-powered applications in critical thinking (CT) development for English as a Foreign Language (EFL) teaching and learning from 2022 to 2025. Firstly, AI-generative tools and LLMs possess both the potential to nurture and the risk of hindering CT in EFL education. 66.67% of studies reported generative AI tools and LLMs' positive role in CT, mainly through improving argumentative speaking performance and writing quality, fostering analytical thinking, reasoning, evaluation, decision-making, and problem-solving skills, enhancing creativity and collaboration, increasing engagement, motivation, and confidence, and supporting self-directed and personalized learning. While 33.33% of studies reported its negative role in CT, highlighting the risk of overreliance on generative AI and LLMs, leading to passive thinking, declining analytical depth, impeding autonomous skill development, and reducing critical thinking and creativity. Their impact largely depends on how they are integrated into EFL educational practices. When used strategically, through guided learning, critical engagement, and AI literacy training, generative AI and LLMs can serve as powerful tools to enhance EFL learners' CT. Therefore, a balanced, well-structured approach is essential to harness generative AI's benefits while fostering independent thought and deep intellectual engagement.

Moreover, in this systematic review, 3 types of research methods, qualitative study, quantitative study, and mixed methods, were all found in the field. Furthermore, the main research focuses on generative AI tools and LLMs in CT development within EFL education revolved 3 key themes: the impact of generative AI and LLMs on EFL

students' CT skills, generative AI's role in EFL teaching and learning, and the influence of generative AI on EFL teachers' competencies and instructional practices. In addition, 4 groups of theoretical perspectives on this field highlighted their potential to enhance CT in EFL education through cognitive theories, sociocultural theories, critical thinking and argumentation frameworks, and AI-based learning models. As Generative AI and LLMs continue to evolve, these theoretical insights are robust foundations for guiding their effective integration into EFL learners' CT development.

Finally, while these generative AI technologies offer new ways to foster EFL learners' CT in EFL teaching and learning, their limitations highlight concerns that need to be addressed in future research. Issues related to research scope, user dependency, generative AI reliability, and pedagogical integration should be systematically addressed. These limitations highlight the constraints of the current research and suggest areas for further investigation. Future research on generative AI tools and LLMs on CT in EFL education should focus on assessing long-term effects, broadening research scope, promoting responsible AI use, and refining pedagogical strategies. Addressing these areas helps ensure that AI-generative tools and LLMs-powered tools effectively support the development of EFL learners' CT development.

### **Implications**

This systematic review offers important implications concerning the use of generative AI and LLMs in fostering critical thinking (CT) among English as a Foreign Language (EFL) learners. Theoretically, this study deepens the understanding of the relationship between generative AI, LLMs, and CT development in EFL setting by identifying four core theoretical perspectives: cognitive theories, sociocultural theories, critical thinking and argumentation frameworks, and AI-based learning models. These frameworks reflect the dual impact of AI tools: on the one hand, enhancing learners' reasoning, problem-solving, and creativity; on the other, posing risks of cognitive passivity and overdependence. The influence of AI on CT appears to be closely tied to its pedagogical integration within EFL instruction. Consequently, there is a growing need for more comprehensive or hybrid theoretical models that consider both the affordances and limitations of AI in education. The synthesized findings offer a conceptual basis for further theoretical advancement in this emerging field.

Practically, this systematic review provides actionable implications for EFL instruction. When implemented with clear pedagogical aims and appropriate scaffolding, generative AI and LLMs have the potential to enhance learners' CT skills, particularly in areas such as argumentative writing and speaking, analytical reasoning, and learner motivation. Effective use requires structured guidance, critical reflection, and explicit instruction in AI literacy, enabling learners to interact actively with AI-generated content rather than accept it uncritically. AI integration in classroom settings should aim to support, rather than substitute, learners' cognitive efforts. Such practices can foster greater learner autonomy and deeper intellectual engagement in EFL contexts.

From a policy perspective, this study calls for the development of clear, forward-thinking guidelines to ensure responsible, ethical, and pedagogically sound use of generative AI in EFL education. Priority should be given to sustained professional development programs that equip educators with the necessary skills and knowledge for AI integration.

Additionally, curriculum reforms should incorporate AI-supported CT tasks that promote critical dialogue, evaluation, and creativity. Policy considerations must also address challenges such as learner overreliance, AI reliability, digital inequality, data privacy, and potential cognitive deskilling. Addressing these issues can help align AI use with long-term educational goals and foster meaningful CT development among EFL learners.

### **Limitations**

While this systematic review provides valuable insights into the role of generative Artificial Intelligence (AI) tools and Large Language Models (LLMs) in fostering critical thinking (CT) in English as a Foreign Language (EFL) teaching and learning, due to its dual nature, research methods, main focuses, theory and models, limitations and future directions of 15 relevant studies, several limitations need be acknowledged in this study.

Firstly, the inclusion criteria of this systematic review focused specifically on generative AI tools and LLMs like ChatGPT, excluding other types of AI tools that may also contribute to fostering CT in EFL education. During the selection process, ten studies on broader AI tools were excluded to maintain the study's scope. However, after completing the review, it became evident that a broader investigation into various AI technologies and their impact on CT in EFL learning could be a valuable direction for future research.

Furthermore, due to the novelty of generative AI tools and LLMs in fostering CT in a specific discipline like EFL course, the available literature is relatively scarce. Consequently, the selected studies included not only peer-reviewed journal articles but also one conference paper and three academic dissertations. While these sources contribute valuable insights, they may lack the rigorous validation processes typical of fully peer-reviewed publications. This limitation suggests the need for further empirical studies and peer-reviewed research in this area to strengthen the evidence base.

Despite these limitations, this review serves as a foundational step in understanding the intersection of generative AI, LLMs, and CT in EFL education, highlighting areas for further exploration and refinement in future research.

### **Future directions**

Based on the limitations identified in this systematic review, future research can explore two key areas to expand our understanding of how Artificial Intelligence (AI) tools, particularly Generative AI and Large Language Models (LLMs), support critical thinking (CT) in EFL education.

Firstly, future studies can consider a wider range of AI tools beyond generative AI and LLMs-powered applications. This review focused on a specific subset of AI applications, excluding other technologies such as adaptive learning systems, automated feedback tools, and AI-driven tutoring platforms. These tools may also contribute to developing EFL students' CT skills in different ways. Expanding the research to include these technologies would allow for a more complete comparison of their effectiveness and potential synergies. Understanding the unique contributions of various AI tools could help EFL educators make more informed decisions about technology integration in the EFL classroom.

Moreover, given the limited number of peer-reviewed studies in this area, future research should emphasize rigorous empirical investigations. The current systematic review includes conference paper and academic dissertations, which, while valuable, may lack the validation process of fully peer-reviewed journal articles. Conducting longitudinal studies and controlled experiments could strengthen the evidence base and provide more reliable insights into how generative AI tools and LLMs influence EFL learners' CT development over time. Encouraging more peer-reviewed publications in this field will enhance the credibility of findings and support the development of best practices for generative AI-assisted learning.

### Appendix 1 Research focuses and limitations of 15 selected articles in systematic literature review

| No. | Authors (year)            | Regions   | Study design       | Main focus                                                                                                                                              | Limitations                                                                                                                                                                                                  |
|-----|---------------------------|-----------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Avsheniuk et al. (2024)   | Ukraine   | Mixed methods      | The study focuses on investigating how AI tools like ChatGPT influence students' critical thinking development in language learning environments.       | Small sample size; short study duration; potential cultural specificity of findings                                                                                                                          |
| 2   | Aymen and Zakarya (2024)  | Algeria   | Mixed methods      | The study focuses on the impact of AI tools on students' critical thinking skills.                                                                      | Limited sample size; focus on a specific university; potential bias in self-reported data                                                                                                                    |
| 3   | Darmawansah et al. (2024) | Taiwan    | Quantitative Study | The study focuses on that ChatGPT guides students through collaboration script stages to enhance argumentation skills.                                  | Prompt quality affects ChatGPT effectiveness                                                                                                                                                                 |
| 4   | Datskiv et al. (2024)     | Ukraine   | Qualitative study  | The study focuses on developing future teachers' academic writing and critical thinking skills using ChatGPT.                                           | ChatGPT has limitations in content generation, potential misinformation, and inability to differentiate reliable information; students and educators must verify information and maintain academic integrity |
| 5   | Hading et al. (2024)      | Indonesia | Qualitative study  | The study focuses on analyzing students' perceptions of AI's impact on their critical thinking abilities, including both positive and negative effects. | Limited scope (one institution); limited perspective (only student views); qualitative impact measurement; rapid technological development                                                                   |

| No. | Authors (year)          | Regions     | Study design       | Main focus                                                                                                                                                                                | Limitations                                                                                                                     |
|-----|-------------------------|-------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 6   | Hapsari and Wu (2022)   | Taiwan      | Qualitative study  | Investgating AI Chat-bots Learning Model in English Speaking Skill to Alleviate speaking anxiety, boosting enjoyment, and fostering critical thinking.                                    | Relatively small sample size                                                                                                    |
| 7   | Khan (2024)             | Pakistan    | Qualitative study  | The study focuses on the AI tools' impact on critical thinking, creativity, and intellectual independence in students.                                                                    | Limited to two universities in Pakistan; focused on BS English program                                                          |
| 8   | Lee and Davis (2024)    | South Korea | Quantitative study | The study focuses on the effects of generative AI on affective factors and the use of generative AI in writing tasks encourage critical thinking.                                         | Short timeframe (one semester); not delve deeper into learners' experiences                                                     |
| 9   | Liu and Wang (2024)     | China       | Mixed methods      | The study focuses on enhancing critical thinking abilities through AI tools like ChatGPT and SummarizBot.                                                                                 | Reliance on self-reported measures of critical thinking; relatively small sample size; limited to a specific context            |
| 10  | Nabilla et al. (2024)   | Indonesia   | Qualitative study  | The study focuses on application of ParagraphAI Technology in educational process for writing skills and critical thinking development.                                                   | Students should not rely solely on technology; ParagraphAI should be used as a tool for learning and writing, not a replacement |
| 11  | Octavio et al. (2024)   | Spain       | Qualitative study  | The study highlights the need for EFL teachers to possess critical thinking skills alongside AI competencies to recognize inaccuracies and identify relevant outcomes when using ChatGPT. | Single case study; free version of ChatGPT (GPT-3) has limitations compared to newer versions.                                  |
| 12  | Rizkiani et al. (2024)  | Indonesia   | Qualitative study  | The study focuses on the influence of AI tools on developing critical thinking abilities in EFL writing.                                                                                  | Overreliance on AI may impede the development of autonomous analytical skills                                                   |
| 13  | Xhaferi and Musa (2024) | Kosovo      | Mixed methods      | The study focuses on enhancing critical thinking skills in English language learning.                                                                                                     | The research topic was relatively new, which might have affected the depth and breadth of the study.                            |
| 14  | Yilin et al. (2023)     | Malaysia    | Qualitative study  | The study focuses on enhancing students' critical thinking skills through ChatGPT integration in EFL teaching.                                                                            | Limited number of participants; teachers are still in the initial exploring stage of using ChatGPT                              |

| No. | Authors (year)      | Regions | Study design  | Main focus                                                                                                                            | Limitations                                                            |
|-----|---------------------|---------|---------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 15  | Zhang et al. (2024) | China   | Mixed methods | The study focuses on how AI chatbots can enhance students' critical thinking skills in the process of writing English speech scripts. | The lack of targeted guidance on leveraging the AI chatbot effectively |

Author contributions

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Data availability

Data generated or analyzed during this study are available from the authors on request.

Declarations

Ethics approval and consent to participate

We make sure that no ethics approval issue involved in the article. All the participants attended the study voluntarily and were rewarded with gifts.

Competing interests

The authors declare no competing interest.

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